

Roll No. : _____

CHARTERED INSTITUTE OF TECHNOLOGY, ABU ROAD

B.Tech. I Sem Examination, 2023

1FY2-09 EME

10:00 TO 11:30 AM

DATE: 07.11.2023

Maximum Marks: 40

1 Year / 1 Sem

STAFF NAME: - Mohsin Ali

Minimum Marks: 20

SECTION A

10

Short Answer Questions – (each of 2 marks)

1. What is meant by closed system? Give an example.
2. Explain briefly zeroth law of thermodynamics
3. What do you mean of latent heating and latent cooling?
4. What is meant by system, surroundings and boundary?
5. What is meant by thermodynamic equilibrium?

SECTION B

16

Long Answer based Questions (any TWO) – (each of 8 marks)

1. Explain Power producing Rankine cycle in details.
2. Explain about Carnot cycle and explain why it's not practically impossible
3. Explain first law of thermodynamics in details.

SECTION C

14

Long Answer based Questions (any 1) – (each of 14 marks)

1. Describe with neat sketches working of a simple vapour compression Refrigeration system
2. Calculate the non-flow Work done for a gas which expands from initial pressure 10 bar and volume 5 m^3 to final volume 20 m^3 under following reversible process. Show the process P-V diagram. Polytrophic index $n=1.3$

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ABU ROAD INSTITUTE OF TECHNOLOGY, ABU ROAD
B.Tech. I Sem II Internal Examination, 2023-24
2FY2-09: ELEMENT OF MECHANICAL ENGINEERING
1st Year / 1st Sem
10:00 AM TO 11:45 AM

STAFF NAME:- TUSHAR BAROT

DATE: 30.01.2024
Maximum Marks: 60
Minimum Marks: 30

SECTION -A

Short Answer Questions - (attempt all, each of 2 marks)

1. What do you mean by materials?
2. What do you mean by Hydrogels?
3. Write short note on Piezoelectric Materials?
4. Write short note on Ductility?
5. Write a short note on Shape Memory Alloys?

SECTION-B

Medium Answer based Questions - (Any Two, each of 10 marks)

1. Explain in detail the metal forming process?
2. Describe various material alloys and their applications?
3. Explain in detail internal combustion engine with classifications?

SECTION-C

Long Answer based Questions - (Any TWO, each of 15 marks)

- Q1. Drive the ratio of tension in belt drive?
- Q2. Gives a detail Materials can be classified into various categories based on different criteria?
- Q3. A gas turbine plant operates on the Brayton cycle between $T_{min} = 300 \text{ K}$ and $T_{max} = 1073 \text{ K}$. find the maximum work done per kg of air and the corresponding cycle efficiency. How does this efficiency compare with the Carnot cycle efficiency operating between the same two temperatures?

27 HT $\rightarrow R_n$
 $= U R_n$

Section A

Short Answer Questions – (each of 2 marks)

1. Explain second law of thermodynamics
2. Explain first law of thermodynamics
3. What is mean of specific heat?
4. Draw Carnot cycle in T-S diagram.
5. Define the term thermal engineering
6. What is meant by closed system? Give an example.
7. Define: Specific heat capacity at constant pressure.
8. What is meant by system, surroundings and boundary?
9. What is meant by thermodynamic equilibrium?
10. Define cold working metal

20

Section B

Long Answer based Questions – (each of 4 marks) ANY 5

20

Q.1 An irreversible heat engine extracts heat from a high – temperature source at a rate of 100 kW and rejects heat to a sink at a rate of 50 kW. The entire work output of the heat engine is used to drive a reversible heat pump operating between a set of independent isothermal heat reservoirs at 17°C and 75°C. What is the rate (in kW) at which the heat pump delivers heat to its high-temperature sink?

Q.2 Differentiate between two-stroke and four-stroke I.C. engines.

Q.3 Explain turbocharging and supercharging in S.I. engines.

Q.4 Mention the various processes in the diesel cycle and dual cycle.

Q.5 Why is the Brayton cycle used in gas turbines?

Q.6 Explain the gas welding process.

Q.7 Write short notes on ductility, annealing and toughness.

SECTION - C

Long Answer based Questions - (each of 10 marks) ANY 3

730

Q.1 Explain rankin cycle with diagram

Q.2 A Carnot refrigerator system has a working temperature of -30°C and 40°C . What is the maximum C.O.P. possible? If the actual C.O.P. is 75% of maximum, calculate the actual refrigeration effect produced per kWh and the capacity of the system.

Q.3 Estimate the loss in air standard efficiency for the diesel engine for the compression ratio 14 and the cutoff changes from 6% to 13% of the stroke.

Q.4 Discuss the various metal forming processes in detail.

Q.5 An engine running at 150 r.p.m. drives a line shaft by means of a belt. The engine pulley is 750 mm in diameter and the pulley on the line shaft is 450 mm. A 900 mm diameter pulley on the line shaft drives a 150 mm diameter pulley keyed to a dynamo shaft. Find the speed of the dynamo shaft when;

1. there is no slip, and

2. there is a slip of 2% at each drive.